

# 05-examples

Mohammad Rehan

2520030066

## 1. INNER JOIN

The screenshot shows a database IDE with the following components:

- SQL Editor:** Contains the following SQL code:

```
29 (2,'Surgery',7),
30 (3,'Psychiatry',9);
31
32 • SELECT d.name AS Department,
33        p.name AS Head_Physician
34 FROM department d
35 JOIN physician p
36 ON d.head = p.employeeid;
```
- Result Grid:** Displays the query results in a table:

| Department       | Head_Physician |
|------------------|----------------|
| General Medicine | Pratik Kumar   |
| Surgery          | Arnav Chitkena |
| Psychiatry       | Molly Clock    |
- Action Output:** A log of database actions and their results:

| # | Time     | Action                                                                                                             | Message                      |
|---|----------|--------------------------------------------------------------------------------------------------------------------|------------------------------|
| 1 | 22:03:05 | USE bookflow_db                                                                                                    | 0 row(s) affected            |
| 2 | 22:03:08 | CREATE TABLE physician ( employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR...                        | 0 row(s) affected            |
| 3 | 22:04:11 | INSERT INTO physician VALUES (1,'Mohammad Rehan','Staff Internist','1111111111'), (2,'Akhil Karuturi','Attendin... | 9 row(s) affected Records: 1 |
| 4 | 22:04:14 | CREATE TABLE department ( departmentid INT PRIMARY KEY, name VARCHAR(100), head INT )                              | 0 row(s) affected            |
| 5 | 22:04:16 | INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)                        | 3 row(s) affected Records: 1 |
| 6 | 22:09:25 | SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON d....                  | 3 row(s) returned            |

## 2. WHERE + LIKE

```
33      p.name AS Head_Physician
34  FROM department d
35  JOIN physician p
36  ON d.head = p.employeeid;
37
38  • SELECT *
39  FROM physician
40  WHERE position = 'Surgical Attending Physician';
41
```

| employeeid | name             | position                     | ssn       |
|------------|------------------|------------------------------|-----------|
| 3          | Rohit Upputuri   | Surgical Attending Physician | 333333333 |
| 6          | Sreekar Varanasi | Surgical Attending Physician | 666666666 |
| 7          | Arnav Chitkena   | Surgical Attending Physician | 777777777 |
| NULL       | NULL             | NULL                         | NULL      |

physician 2 x

Output

Action Output

| # | Time     | Action                                                                                                          | Message                 |
|---|----------|-----------------------------------------------------------------------------------------------------------------|-------------------------|
| 2 | 22:03:08 | CREATE TABLE physician ( employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR...                     | 0 row(s) affected       |
| 3 | 22:04:11 | INSERT INTO physician VALUES (1,'Mohammad Rehan','Staff Internist','111111111'), (2,'Akhlil Karuturi','Atten... | 9 row(s) affected Recon |
| 4 | 22:04:14 | CREATE TABLE department ( departmentid INT PRIMARY KEY, name VARCHAR(100), head INT )                           | 0 row(s) affected       |
| 5 | 22:04:16 | INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)                     | 3 row(s) affected Recon |
| 6 | 22:09:25 | SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...                 | 3 row(s) returned       |
| 7 | 22:26:40 | SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000                           | 3 row(s) returned       |

```
37
38  • SELECT *
39  FROM physician
40  WHERE position = 'Surgical Attending Physician';
41
42  • SELECT *
43  FROM physician
44  WHERE name LIKE 'John%';
```

| employeeid | name | position | ssn  |
|------------|------|----------|------|
| NULL       | NULL | NULL     | NULL |

physician 3 x

Output

Action Output

| # | Time     | Action                                                                                                          | Message                    |
|---|----------|-----------------------------------------------------------------------------------------------------------------|----------------------------|
| 3 | 22:04:11 | INSERT INTO physician VALUES (1,'Mohammad Rehan','Staff Internist','111111111'), (2,'Akhlil Karuturi','Atten... | 9 row(s) affected Records: |
| 4 | 22:04:14 | CREATE TABLE department ( departmentid INT PRIMARY KEY, name VARCHAR(100), head INT )                           | 0 row(s) affected          |
| 5 | 22:04:16 | INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)                     | 3 row(s) affected Records: |
| 6 | 22:09:25 | SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...                 | 3 row(s) returned          |
| 7 | 22:26:40 | SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000                           | 3 row(s) returned          |
| 8 | 22:27:08 | SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000                                                   | 0 row(s) returned          |

### 3. COUNT + COUNT(DISTINCT)

DML DDL DCL-TCL 03 04-joins-set-operations 05-examples\*

Limit to 1000 rows

```
40 WHERE position = 'Surgical Attending Physician';
41
42 • SELECT *
43 FROM physician
44 WHERE name LIKE 'John%';
45
46 • SELECT COUNT(*) AS Total_Physicians
47 FROM physician;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

|   | Total_Physicians |
|---|------------------|
| ▶ | 9                |

Result 4 x

Output

Action Output

| # | Time       | Action                                                                                          | Message                   |
|---|------------|-------------------------------------------------------------------------------------------------|---------------------------|
| ✓ | 4 22:04:14 | CREATE TABLE department ( departmentid INT PRIMARY KEY, name VARCHAR(100), head INT )           | 0 row(s) affected         |
| ✓ | 5 22:04:16 | INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)     | 3 row(s) affected Records |
| ✓ | 6 22:09:25 | SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ... | 3 row(s) returned         |
| ✓ | 7 22:26:40 | SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000           | 3 row(s) returned         |
| ✓ | 8 22:27:08 | SELECT * FROM physician WHERE name LIKE 'John%'; LIMIT 0, 1000                                  | 0 row(s) returned         |
| ✓ | 9 22:28:37 | SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000                                | 1 row(s) returned         |

DML DDL DCL-TCL 03 04-joins-set-operations 05-examples\*

Limit to 1000 rows

```
43 FROM physician
44 WHERE name LIKE 'John%';
45
46 • SELECT COUNT(*) AS Total_Physicians
47 FROM physician;
48
49 • SELECT COUNT(DISTINCT position) AS Different_Positions
50 FROM physician;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

|   | Different_Positions |
|---|---------------------|
| ▶ | 7                   |

Result 5 x

Output

Action Output

| # | Time        | Action                                                                                          | Message           |
|---|-------------|-------------------------------------------------------------------------------------------------|-------------------|
| ✓ | 5 22:04:16  | INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)     | 3 row(s) affected |
| ✓ | 6 22:09:25  | SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ... | 3 row(s) returned |
| ✓ | 7 22:26:40  | SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000           | 3 row(s) returned |
| ✓ | 8 22:27:08  | SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000                                   | 0 row(s) returned |
| ✓ | 9 22:28:37  | SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000                                | 1 row(s) returned |
| ✓ | 10 22:28:48 | SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000             | 1 row(s) returned |

## 4. GROUP BY + HAVING

DML DDL DCL-TCL 03 04-joins-set-operations 05-examples\*

Limit to 1000 rows

```
48
49 • SELECT COUNT(DISTINCT position) AS Different_Positions
50 FROM physician;
51
52 • SELECT position,
53       COUNT(*) AS Total_Employees
54 FROM physician
55 GROUP BY position;
```

Result Grid

| position                     | Total_Employees |
|------------------------------|-----------------|
| Staff Internist              | 1               |
| Attending Physician          | 1               |
| Surgical Attending Physician | 3               |
| Senior Attending Physician   | 1               |
| Head Chief of Medicine       | 1               |
| MD Resident                  | 1               |
| Attending Psychiatrist       | 1               |

Result 6

Output

Action Output

| #  | Time     | Action                                                                                          | Message           |
|----|----------|-------------------------------------------------------------------------------------------------|-------------------|
| 6  | 22:09:25 | SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ... | 3 row(s) returned |
| 7  | 22:26:40 | SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000           | 3 row(s) returned |
| 8  | 22:27:08 | SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000                                   | 0 row(s) returned |
| 9  | 22:28:37 | SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000                                | 1 row(s) returned |
| 10 | 22:28:48 | SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000             | 1 row(s) returned |
| 11 | 22:30:28 | SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000     | 7 row(s) returned |

```
57 • SELECT position,
58       COUNT(*) AS Total_Employees
59 FROM physician
60 GROUP BY position
61 HAVING COUNT(*) > 1;
```

Result Grid

| position                     | Total_Employees |
|------------------------------|-----------------|
| Surgical Attending Physician | 3               |

Result 7

Output

Action Output

| #  | Time     | Action                                                                                        | Message           |
|----|----------|-----------------------------------------------------------------------------------------------|-------------------|
| 7  | 22:26:40 | SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000         | 3 row(s) returned |
| 8  | 22:27:08 | SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000                                 | 0 row(s) returned |
| 9  | 22:28:37 | SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000                              | 1 row(s) returned |
| 10 | 22:28:48 | SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000           | 1 row(s) returned |
| 11 | 22:30:28 | SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000   | 7 row(s) returned |
| 12 | 22:30:50 | SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT... | 1 row(s) returned |

## 5. ORDER BY

```
63 • SELECT *
64 FROM physician
65 ORDER BY name ASC;
66
```

Result Grid Filter Rows:  Edit: Export/Import: Wrap Cell Content:

|   | employeeid | name              | position                     | ssn      |
|---|------------|-------------------|------------------------------|----------|
| ▶ | 2          | Akhil Karuturi    | Attending Physician          | 22222222 |
|   | 7          | Arnav Chitkena    | Surgical Attending Physician | 77777777 |
|   | 8          | Keith Dudemeister | MD Resident                  | 88888888 |
|   | 1          | Mohammad Rehan    | Staff Internist              | 11111111 |
|   | 9          | Molly Clock       | Attending Psychiatrist       | 99999999 |
|   | 4          | Pratik Kumar      | Senior Attending Physician   | 44444444 |
|   | 3          | Rohit Upputuri    | Surgical Attending Physician | 33333333 |
|   | 6          | Sreekar Varanasi  | Surgical Attending Physician | 66666666 |
|   | 5          | Yugandar Peddi    | Head Chief of Medicine       | 55555555 |
| • | NULL       | NULL              | NULL                         | NULL     |

physician 8 x

Output

Action Output

| # | Time        | Action                                                                                        | Message           |
|---|-------------|-----------------------------------------------------------------------------------------------|-------------------|
| ✓ | 8 22:27:08  | SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000                                 | 0 row(s) returned |
| ✓ | 9 22:28:37  | SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000                              | 1 row(s) returned |
| ✓ | 10 22:28:48 | SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000           | 1 row(s) returned |
| ✓ | 11 22:30:28 | SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000   | 7 row(s) returned |
| ✓ | 12 22:30:50 | SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT... | 1 row(s) returned |
| ✓ | 13 22:31:34 | SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000                                       | 9 row(s) returned |

```
67 • SELECT *
68 FROM physician
69 ORDER BY employeeid DESC;
70
```

Result Grid Filter Rows:  Edit: Export/Import: Wrap Cell Content:

|   | employeeid | name              | position                     | ssn      |
|---|------------|-------------------|------------------------------|----------|
| ▶ | 9          | Molly Clock       | Attending Psychiatrist       | 99999999 |
|   | 8          | Keith Dudemeister | MD Resident                  | 88888888 |
|   | 7          | Arnav Chitkena    | Surgical Attending Physician | 77777777 |
|   | 6          | Sreekar Varanasi  | Surgical Attending Physician | 66666666 |
|   | 5          | Yugandar Peddi    | Head Chief of Medicine       | 55555555 |
|   | 4          | Pratik Kumar      | Senior Attending Physician   | 44444444 |
|   | 3          | Rohit Upputuri    | Surgical Attending Physician | 33333333 |
|   | 2          | Akhil Karuturi    | Attending Physician          | 22222222 |
|   | 1          | Mohammad Rehan    | Staff Internist              | 11111111 |
| • | NULL       | NULL              | NULL                         | NULL     |

physician 9 x

Output

Action Output

| # | Time        | Action                                                                                        | Message           |
|---|-------------|-----------------------------------------------------------------------------------------------|-------------------|
| ✓ | 9 22:28:37  | SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000                              | 1 row(s) returned |
| ✓ | 10 22:28:48 | SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000           | 1 row(s) returned |
| ✓ | 11 22:30:28 | SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000   | 7 row(s) returned |
| ✓ | 12 22:30:50 | SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT... | 1 row(s) returned |
| ✓ | 13 22:31:34 | SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000                                       | 9 row(s) returned |
| ✓ | 14 22:32:04 | SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000                                | 9 row(s) returned |



## 6. SUBQUERIES — IN + NOT IN

```
71 • SELECT *
72 FROM physician
73 WHERE employeeid IN
74 (
75     SELECT head
76     FROM department
77 );
78
```

| Result Grid | Filter Rows:   | Edit:                        | Export/Import: | Wrap Cell Content: |
|-------------|----------------|------------------------------|----------------|--------------------|
| employeeid  | name           | position                     | ssn            |                    |
| 4           | Prabk Kumar    | Senior Attending Physician   | 444444444      |                    |
| 7           | Arnav Chitkena | Surgical Attending Physician | 777777777      |                    |
| 9           | Molly Clock    | Attending Psychiatrist       | 999999999      |                    |
| NULL        | NULL           | NULL                         | NULL           |                    |

physician 10 x

Output

Action Output

| #    | Time     | Action                                                                                        | Message           |
|------|----------|-----------------------------------------------------------------------------------------------|-------------------|
| ✓ 10 | 22:28:48 | SELECT COUNT(DISTINCT position) AS Different_Positions FROM physician LIMIT 0, 1000           | 1 row(s) returned |
| ✓ 11 | 22:30:28 | SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000   | 7 row(s) returned |
| ✓ 12 | 22:30:50 | SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT... | 1 row(s) returned |
| ✓ 13 | 22:31:34 | SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000                                       | 9 row(s) returned |
| ✓ 14 | 22:32:04 | SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000                                | 9 row(s) returned |
| ✓ 15 | 22:33:05 | SELECT * FROM physician WHERE employeeid IN ( SELECT head FROM department ) LIMIT 0, 1000     | 3 row(s) returned |

```
79 • SELECT *
80 FROM physician
81 WHERE employeeid NOT IN
82 (
83     SELECT head
84     FROM department
85 );
86
```

| Result Grid | Filter Rows:      | Edit:                        | Export/Import: | Wrap Cell Content: |
|-------------|-------------------|------------------------------|----------------|--------------------|
| employeeid  | name              | position                     | ssn            |                    |
| 1           | Mohammad Rehan    | Staff Internist              | 111111111      |                    |
| 2           | Akhil Karuturi    | Attending Physician          | 222222222      |                    |
| 3           | Rohit Upputuri    | Surgical Attending Physician | 333333333      |                    |
| 5           | Yugandar Peddi    | Head Chief of Medicine       | 555555555      |                    |
| 6           | Sreekar Varanasi  | Surgical Attending Physician | 666666666      |                    |
| 8           | Keith Dudemeister | MD Resident                  | 888888888      |                    |
| NULL        | NULL              | NULL                         | NULL           |                    |

physician 11 x

Output

Action Output

| #    | Time     | Action                                                                                        | Message           |
|------|----------|-----------------------------------------------------------------------------------------------|-------------------|
| ✓ 11 | 22:30:28 | SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000   | 7 row(s) returned |
| ✓ 12 | 22:30:50 | SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT... | 1 row(s) returned |
| ✓ 13 | 22:31:34 | SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000                                       | 9 row(s) returned |
| ✓ 14 | 22:32:04 | SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000                                | 9 row(s) returned |
| ✓ 15 | 22:33:05 | SELECT * FROM physician WHERE employeeid IN ( SELECT head FROM department ) LIMIT 0, 1000     | 3 row(s) returned |
| ✓ 16 | 22:33:22 | SELECT * FROM physician WHERE employeeid NOT IN ( SELECT head FROM department ) LIMIT 0, 1... | 6 row(s) returned |

## 7. GROUP BY + ORDER BY + LIKE

```
87 • SELECT position,  
88     COUNT(*) AS Total_Employees  
89 FROM physician  
90 GROUP BY position  
91 ORDER BY Total_Employees DESC;  
92
```

| Result Grid                  |                 | Filter Rows: | Export: | Wrap Cell Content: |
|------------------------------|-----------------|--------------|---------|--------------------|
| position                     | Total_Employees |              |         |                    |
| Surgical Attending Physician | 3               |              |         |                    |
| Staff Internist              | 1               |              |         |                    |
| Attending Physician          | 1               |              |         |                    |
| Senior Attending Physician   | 1               |              |         |                    |
| Head Chief of Medicine       | 1               |              |         |                    |
| MD Resident                  | 1               |              |         |                    |
| Attending Psychiatrist       | 1               |              |         |                    |

Result 12 x

Output

Action Output

| #    | Time     | Action                                                                                          | Message           |
|------|----------|-------------------------------------------------------------------------------------------------|-------------------|
| ✓ 12 | 22:30:50 | SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...   | 1 row(s) returned |
| ✓ 13 | 22:31:34 | SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000                                         | 9 row(s) returned |
| ✓ 14 | 22:32:04 | SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000                                  | 9 row(s) returned |
| ✓ 15 | 22:33:05 | SELECT * FROM physician WHERE employeeid IN ( SELECT head FROM department ) LIMIT 0, 1000       | 3 row(s) returned |
| ✓ 16 | 22:33:22 | SELECT * FROM physician WHERE employeeid NOT IN ( SELECT head FROM department ) LIMIT 0, 1...   | 6 row(s) returned |
| ✓ 17 | 22:34:00 | SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total... | 7 row(s) returned |
| 93   |          | SELECT *                                                                                        |                   |
| 94   |          | FROM physician                                                                                  |                   |
| 95   |          | WHERE position LIKE '%Attending%';                                                              |                   |
| 96   |          |                                                                                                 |                   |

| Result Grid |                  | Filter Rows:                 | Edit:     | Export/Import: | Wrap Cell Content: |
|-------------|------------------|------------------------------|-----------|----------------|--------------------|
| employeeid  | name             | position                     | ssn       |                |                    |
| 2           | Akhil Karuturi   | Attending Physician          | 222222222 |                |                    |
| 3           | Rohit Upputuri   | Surgical Attending Physician | 333333333 |                |                    |
| 4           | Pratik Kumar     | Senior Attending Physician   | 444444444 |                |                    |
| 6           | Sreekar Varanasi | Surgical Attending Physician | 666666666 |                |                    |
| 7           | Arnav Chitkena   | Surgical Attending Physician | 777777777 |                |                    |
| 9           | Molly Clock      | Attending Psychiatrist       | 999999999 |                |                    |
| NULL        | NULL             | NULL                         | NULL      |                |                    |

physician 13 x

Output

Action Output

| #    | Time     | Action                                                                                          | Message           |
|------|----------|-------------------------------------------------------------------------------------------------|-------------------|
| ✓ 13 | 22:31:34 | SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000                                         | 9 row(s) returned |
| ✓ 14 | 22:32:04 | SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000                                  | 9 row(s) returned |
| ✓ 15 | 22:33:05 | SELECT * FROM physician WHERE employeeid IN ( SELECT head FROM department ) LIMIT 0, 1000       | 3 row(s) returned |
| ✓ 16 | 22:33:22 | SELECT * FROM physician WHERE employeeid NOT IN ( SELECT head FROM department ) LIMIT 0, 1...   | 6 row(s) returned |
| ✓ 17 | 22:34:00 | SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total... | 7 row(s) returned |
| ✓ 18 | 22:34:16 | SELECT * FROM physician WHERE position LIKE '%Attending%' LIMIT 0, 1000                         | 6 row(s) returned |

## 8. JOIN + WHERE / JOIN + ORDER BY

```
97 • SELECT p.name
98 FROM physician p
99 JOIN department d
100 ON p.employeeid = d.head
101 WHERE d.name = 'Surgery';
102
```

| Result Grid      | Filter Rows: | Export: | Wrap Cell Content: |
|------------------|--------------|---------|--------------------|
| name             |              |         |                    |
| ▶ Arnav Chitkena |              |         |                    |

Result 14 x

Output

Action Output

| #    | Time     | Action                                                                                               | Message           |
|------|----------|------------------------------------------------------------------------------------------------------|-------------------|
| ✓ 14 | 22:32:04 | SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000                                       | 9 row(s) returned |
| ✓ 15 | 22:33:05 | SELECT * FROM physician WHERE employeeid IN ( SELECT head FROM department ) LIMIT 0, 1000            | 3 row(s) returned |
| ✓ 16 | 22:33:22 | SELECT * FROM physician WHERE employeeid NOT IN ( SELECT head FROM department ) LIMIT 0, 1...        | 6 row(s) returned |
| ✓ 17 | 22:34:00 | SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...      | 7 row(s) returned |
| ✓ 18 | 22:34:16 | SELECT * FROM physician WHERE position LIKE '%Attending%' LIMIT 0, 1000                              | 6 row(s) returned |
| ✓ 19 | 22:34:51 | SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery... | 1 row(s) returned |

```
103 • SELECT d.name,
104         p.name
105 FROM department d
106 JOIN physician p
107 ON d.head = p.employeeid
108 ORDER BY d.name;
109
```

| Result Grid        | Filter Rows:   | Export: | Wrap Cell Content: |
|--------------------|----------------|---------|--------------------|
| name               | name           |         |                    |
| ▶ General Medicine | Pratik Kumar   |         |                    |
| Psychiatry         | Molly Clock    |         |                    |
| Surgery            | Arnav Chitkena |         |                    |

Result 15 x

Output

Action Output

| #    | Time     | Action                                                                                               | Message           |
|------|----------|------------------------------------------------------------------------------------------------------|-------------------|
| ✓ 15 | 22:33:05 | SELECT * FROM physician WHERE employeeid IN ( SELECT head FROM department ) LIMIT 0, 1000            | 3 row(s) returned |
| ✓ 16 | 22:33:22 | SELECT * FROM physician WHERE employeeid NOT IN ( SELECT head FROM department ) LIMIT 0, 1...        | 6 row(s) returned |
| ✓ 17 | 22:34:00 | SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...      | 7 row(s) returned |
| ✓ 18 | 22:34:16 | SELECT * FROM physician WHERE position LIKE '%Attending%' LIMIT 0, 1000                              | 6 row(s) returned |
| ✓ 19 | 22:34:51 | SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery... | 1 row(s) returned |
| ✓ 20 | 22:35:11 | SELECT d.name, p.name FROM department d JOIN physician p ON d.head = p.employeeid ORDER BY d...      | 3 row(s) returned |



## 9. NOT LIKE + COUNT + HAVING

```
110 • SELECT *
111 FROM physician
112 WHERE position NOT LIKE '%Surgical%';
113
```

| Result Grid | Filter Rows:      | Edit:                      | Export/Import: | Wrap Cell Content: |
|-------------|-------------------|----------------------------|----------------|--------------------|
| employeeid  | name              | position                   | ssn            |                    |
| 1           | Mohammad Rehan    | Staff Internist            | 111111111      |                    |
| 2           | Akhil Karuturi    | Attending Physician        | 222222222      |                    |
| 4           | Pratik Kumar      | Senior Attending Physician | 444444444      |                    |
| 5           | Yugandar Peddi    | Head Chief of Medicine     | 555555555      |                    |
| 8           | Keith Dudemeister | MD Resident                | 888888888      |                    |
| 9           | Molly Clock       | Attending Psychiatrist     | 999999999      |                    |
| NULL        | NULL              | NULL                       | NULL           |                    |

physician 16 ×

Output

Action Output

| #    | Time     | Action                                                                                               | Message           |
|------|----------|------------------------------------------------------------------------------------------------------|-------------------|
| ✓ 16 | 22:33:22 | SELECT * FROM physician WHERE employeeid NOT IN ( SELECT head FROM department ) LIMIT 0, 1...        | 6 row(s) returned |
| ✓ 17 | 22:34:00 | SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...      | 7 row(s) returned |
| ✓ 18 | 22:34:16 | SELECT * FROM physician WHERE position LIKE "%Attending%" LIMIT 0, 1000                              | 6 row(s) returned |
| ✓ 19 | 22:34:51 | SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery... | 1 row(s) returned |
| ✓ 20 | 22:35:11 | SELECT d.name, p.name FROM department d JOIN physician p ON d.head = p.employeeid ORDER BY d...      | 3 row(s) returned |
| ✓ 21 | 22:35:40 | SELECT * FROM physician WHERE position NOT LIKE "%Surgical%" LIMIT 0, 1000                           | 6 row(s) returned |

```
113 • SELECT position,
114         COUNT(*) AS Total
115 FROM physician
116 GROUP BY position
117 HAVING COUNT(*) >= 2;
118
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

| position                     | Total |
|------------------------------|-------|
| Surgical Attending Physician | 3     |

Result 17 ×

Output

Action Output

| #    | Time     | Action                                                                                               | Message           |
|------|----------|------------------------------------------------------------------------------------------------------|-------------------|
| ✓ 17 | 22:34:00 | SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total...      | 7 row(s) returned |
| ✓ 18 | 22:34:16 | SELECT * FROM physician WHERE position LIKE "%Attending%" LIMIT 0, 1000                              | 6 row(s) returned |
| ✓ 19 | 22:34:51 | SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery... | 1 row(s) returned |
| ✓ 20 | 22:35:11 | SELECT d.name, p.name FROM department d JOIN physician p ON d.head = p.employeeid ORDER BY d...      | 3 row(s) returned |
| ✓ 21 | 22:35:40 | SELECT * FROM physician WHERE position NOT LIKE "%Surgical%" LIMIT 0, 1000                           | 6 row(s) returned |
| ✓ 22 | 22:35:57 | SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIM...      | 1 row(s) returned |

## 10. EXISTS + MAX + MIN

```

119 • SELECT *
120 FROM department d
121 WHERE EXISTS
122 (
123     SELECT 1
124     FROM physician p
125     WHERE p.employeeid = d.head
126 );
127

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

| departmentid | name             | head |
|--------------|------------------|------|
| 1            | General Medicine | 4    |
| 2            | Surgery          | 7    |
| 3            | Psychiatry       | 9    |
| NULL         | NULL             | NULL |

department 18 x

Output

Action Output

| #    | Time     | Action                                                                                               | Message           |
|------|----------|------------------------------------------------------------------------------------------------------|-------------------|
| ✓ 18 | 22:34:16 | SELECT * FROM physician WHERE position LIKE '%Attending%' LIMIT 0, 1000                              | 6 row(s) returned |
| ✓ 19 | 22:34:51 | SELECT p.name FROM physician p JOIN department d ON p.employeeid = d.head WHERE d.name = 'Surgery... | 1 row(s) returned |
| ✓ 20 | 22:35:11 | SELECT d.name, p.name FROM department d JOIN physician p ON d.head = p.employeeid ORDER BY d...      | 3 row(s) returned |
| ✓ 21 | 22:35:40 | SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000                           | 6 row(s) returned |
| ✓ 22 | 22:35:57 | SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIM...      | 1 row(s) returned |
| ✓ 23 | 22:36:21 | SELECT * FROM department d WHERE EXISTS ( SELECT 1 FROM physician p WHERE p.employeei...             | 3 row(s) returned |

```

128 • SELECT MAX(employeeid) AS Highest_Employee_ID
129 FROM physician;
130

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

| Highest_Employee_ID |
|---------------------|
| 9                   |

Result 20 x

Output

Action Output

| #    | Time     | Action                                                                                          | Message           |
|------|----------|-------------------------------------------------------------------------------------------------|-------------------|
| ✓ 20 | 22:35:11 | SELECT d.name, p.name FROM department d JOIN physician p ON d.head = p.employeeid ORDER BY d... | 3 row(s) returned |
| ✓ 21 | 22:35:40 | SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000                      | 6 row(s) returned |
| ✓ 22 | 22:35:57 | SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIM... | 1 row(s) returned |
| ✓ 23 | 22:36:21 | SELECT * FROM department d WHERE EXISTS ( SELECT 1 FROM physician p WHERE p.employeei...        | 3 row(s) returned |
| ✓ 24 | 22:36:44 | SELECT MAX(employeeid) AS Highest_Employee_ID FROM physician LIMIT 0, 1000                      | 1 row(s) returned |
| ✓ 25 | 22:38:26 | SELECT MAX(employeeid) AS Highest_Employee_ID FROM physician LIMIT 0, 1000                      | 1 row(s) returned |

```

131 • SELECT MIN(employeeid) AS Lowest_Employee_ID
132 FROM physician;
133

```

Result Grid | | Filter Rows:  | Export: | Wrap Cell Content:

|   | Lowest_Employee_ID |
|---|--------------------|
| ▶ | 1                  |

Result 21 x

Output

Action Output ▼

|   | #  | Time     | Action                                                                                                  | Message           |
|---|----|----------|---------------------------------------------------------------------------------------------------------|-------------------|
| ✓ | 21 | 22:35:40 | SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000                              | 6 row(s) returned |
| ✓ | 22 | 22:35:57 | SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIMIT 0, 1000  | 1 row(s) returned |
| ✓ | 23 | 22:36:21 | SELECT * FROM department d WHERE EXISTS ( SELECT 1 FROM physician p WHERE p.employeeid = d.employeeid ) | 3 row(s) returned |
| ✓ | 24 | 22:36:44 | SELECT MAX(employeeid) AS Highest_Employee_ID FROM physician LIMIT 0, 1000                              | 1 row(s) returned |
| ✓ | 25 | 22:38:26 | SELECT MAX(employeeid) AS Highest_Employee_ID FROM physician LIMIT 0, 1000                              | 1 row(s) returned |
| ✓ | 26 | 22:38:36 | SELECT MIN(employeeid) AS Lowest_Employee_ID FROM physician LIMIT 0, 1000                               | 1 row(s) returned |